

Zarif Latif

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EXPERIENCE

Founder
Railo

January 2026 - Present, On-site

- Engineered an Automated Security Remediation Engine using Python, integrating into CI/CD pipelines to deterministically identify and fix vulnerabilities in Python and JavaScript/TypeScript codebases at pull-request time.
- Implemented secure integration workflows leveraging GitHub Actions and Regex-based parsing to scan for security flaws (e.g., hardcoded secrets, insecure libraries) and automatically commit secure alternatives.
- Transitioned the architecture to a multi-tenant SaaS model, leading the product roadmap and technical vision.

Research & Innovation Intern
Bangladesh Computer Council

January 2026 - February 2026, Agargaon, Dhaka

- Translated complex research findings into actionable product features, utilizing Python and the MERN stack to build scalable, user-centric software solutions for government-scale ICT initiatives.
- Configured network devices and managed VLAN/Subnet architectures; executed incident response protocols to identify root causes for outages and implemented hardening strategies.

AI Cohort Member
GCI World 2026 (Matsuo Laboratory, University of Tokyo)

April 2026 - Present

- Selected for a highly competitive global AI initiative focused on advanced machine learning and artificial intelligence applications.

PROJECT

Aetherius
github.com/zariffromlatif/Aetherius · February 2026 - April 2026

- Engineered an autonomous causal world model for event-driven financial risk surveillance, framing global market dynamics as a Partially Observable Markov Decision Process (POMDP).
- Optimized Test-Time Compute inference (GPT-5.4/DeepSeek-R1) for a 24/7 autonomous loop using 24-hour prompt caching and strict KV-tensor prefix matching to reduce operational compute costs by 90%.

EDUCATION

Bachelor's degree, Computer Science
BRAC University

PUBLICATIONS

From Prediction to Prescription: Graph-Enhanced Causal Offline Reinforcement Learning for Customer Retention
Academic Thesis · BRAC University (Undergraduate Thesis)

- Researching and developing a novel framework utilizing Graph Neural Networks (GNNs) and Causal Reinforcement Learning (CRL) for proactive customer retention, utilizing the Open Bandit Dataset.

Optimizing Static Malware Analysis: A Lightweight, Explainable Framework using LightGBM and SHAP
Working Paper · Independent Research Draft

- Extracted Information Gain to identify the top 100 discriminative features, reducing the feature space by 95% and training a "Lite" LightGBM model that cut computational overhead by 48.7% while improving the F1-Score to 0.9776.
- Integrated SHAP (SHapley Additive exPlanations) to provide global and local interpretability, assigning marginal contribution values to features to ensure mathematically consistent model transparency.

Cross-Lingual Emergence of Induction Heads in Bengali Language Models: A Mechanistic Interpretability Study
Working Paper · Independent Research Draft

- Investigating the emergence, architectural positioning, and functional strength of induction heads—the circuits responsible for in-context learning—in morphologically rich, Brahmic-script languages.
- Utilizing the Pythia model suite and TransformerLens to conduct scalable circuit detection and analyze how tokenization fragmentation impacts developmental thresholds compared to English.

Verification-Gated Neuro-Symbolic Program Repair with Bounded SMT-Based Validation of Semantic Sanitization
Working Paper · Independent Research Draft

- Designing a neuro-symbolic repair framework combining quantized LLMs for intent inference with deterministic AST transformations to apply programmatic vulnerability patches.
- Implementing a bounded Z3 SMT solver check to semantically validate that candidate patches successfully neutralize targeted taint flows (CWE-79, CWE-89) within their sink context prior to CI/CD admission.

SKILLS

- AI Safety & Interpretability:** Mechanistic Interpretability, TransformerLens, Induction Heads, Formal Verification, Z3 SMT Solver, Neuro-Symbolic AI.
- Machine Learning & Data Science:** PyTorch, TensorFlow, LightGBM, Scikit-learn, Hugging Face, Graph Neural Networks (GNNs), LLM Prompt Caching.
- Software Engineering:** Python (Advanced), Golang, JavaScript (ES6+), C++, MERN Stack, AST Transformations, System Design.
- Cybersecurity & DevOps:** Automated Remediation, Taint Analysis, Penetration Testing, Docker, Kubernetes (K8s), CI/CD Automation.

Tools & Technologies: Node.js, MySQL, React.js, Bootstrap (Framework), JSON Web Token (JWT), TensorFlow, Python (Programming Language), Kotlin, Microsoft Office, Microsoft Excel

Interpersonal Skills: Team Leadership, Problem Solving, Strategic Thinking, Strategy, Analytical Skills, Communication

